

Model  $x_1, x_2, \dots, x_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$  with  $\sigma$  known & a prior  $\mu \sim N(\mu_0, \sigma_0^2)$

Posterior (up to a constant). By Bayes' rule,

$$p(\mu | x_{1:n}) \propto p(x_{1:n} | \mu) p(\mu) = \left[ \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \right] \times \frac{1}{\sqrt{2\pi}\sigma_0} \exp\left(-\frac{(\mu - \mu_0)^2}{2\sigma_0^2}\right)$$

or

$$L(\mu) = \log p(\mu | x_{1:n}) = -\sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2} - \frac{(\mu - \mu_0)^2}{2\sigma_0^2}$$

Diff wrt  $\mu$  & set to zero:

$$\frac{dL}{d\mu} = \sum_{i=1}^n \frac{x_i - \mu}{\sigma^2} - \frac{\mu - \mu_0}{\sigma_0^2} = 0$$

Multiply thru & collect  $\mu$  terms. Using  $\sum_{i=1}^n x_i = N\bar{x}$ .

$$\begin{aligned} \frac{N\bar{x} - N\mu}{\sigma^2} &= \frac{\mu - \mu_0}{\sigma_0^2} \Rightarrow \mu \left( \frac{N}{\sigma^2} + \frac{1}{\sigma_0^2} \right) \\ &= \frac{N\bar{x}}{\sigma^2} + \frac{\mu_0}{\sigma_0^2} \end{aligned}$$

solve for  $\mu$ :

$$\mu_{\text{MMP}} = \frac{\frac{N}{\sigma^2} \bar{x} + \frac{1}{\sigma_0^2} \mu_0}{\frac{N}{\sigma^2} + \frac{1}{\sigma_0^2}}$$

Multiply num & den by  $\frac{\sigma_0^2 \sigma^2}{N}$  to match the form in the NB.

$$\hat{\mu}_{\text{map}} = \frac{\sigma_0^2}{\sigma_0^2 + \frac{\sigma^2}{n}} \bar{x} + \frac{\frac{\sigma^2}{n}}{\sigma_0^2 + \frac{\sigma^2}{n}} \mu_0$$